Jharia Coalfield

🏭 Introduction

The **Jharia Coalfield** is one of India's most important and oldest coal mining regions, located in Jharkhand. It is primarily operated by Bharat Coking Coal Limited, a subsidiary of Coal India Limited. The coalfield is famous for its **high-grade prime coking coal**, which is essential for steel production.

Mining activities in Jharia began in the late 19th century, making it a historically significant coalfield in India. It plays a crucial role in supplying coal to major steel plants, especially those operated by Steel Authority of India Limited. However, the coalfield is also widely known for its **persistent underground coal fires**, which have been burning for decades and pose serious environmental and safety challenges.

🌍 Geographical Overview – Jharia Coalfield

The Jharia Coalfield is located in the **Dhanbad district** of Jharkhand and forms an important part of the Damodar Valley coalfields.

📍 Location & Setting

- Lies between **23°39′N to 23°48′N latitude** and **86°11′E to 86°27′E longitude**
- Covers an area of approximately **450 sq. km**
- Well connected by rail and road networks through Dhanbad

🌄 Physical & Geological Features

- Part of the **Gondwana coalfields**, rich in high-quality coking coal
- Contains **multiple thick and extensive coal seams**
- Suitable for both **underground and opencast mining methods**

🌊 Nearby Natural Features

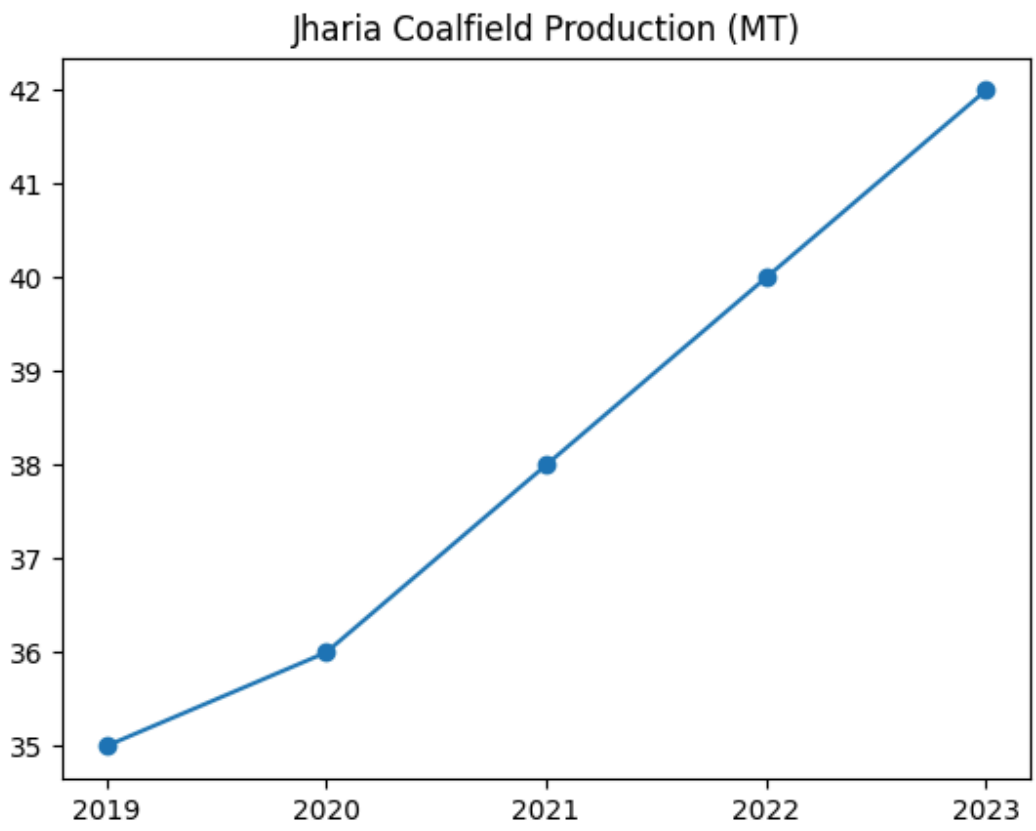
- Located in the **Damodar Valley region**
- Close to the Damodar River, supporting industrial activities

🌦️ Climate

- Tropical climate
- Hot summers (up to ~47°C)

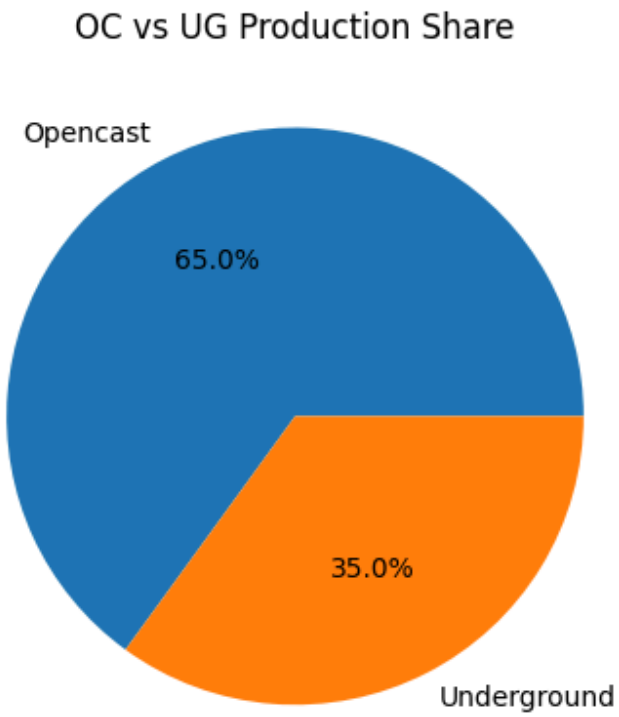
- Moderate monsoon rainfall
- Cool winters

📊 1. Coal Production Trend (MT)



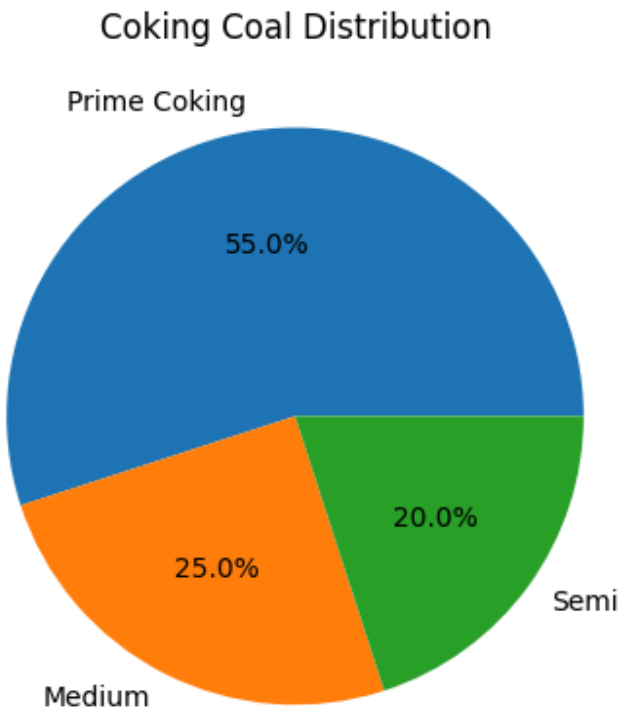
This chart shows the year-wise coal production from Jharia coalfield under Bharat Coking Coal Limited.
Production shows gradual growth despite operational challenges like underground fires.
Significance: Helps in production forecasting and evaluating the performance of a complex mining region.

📊 2. OC vs UG Production Share



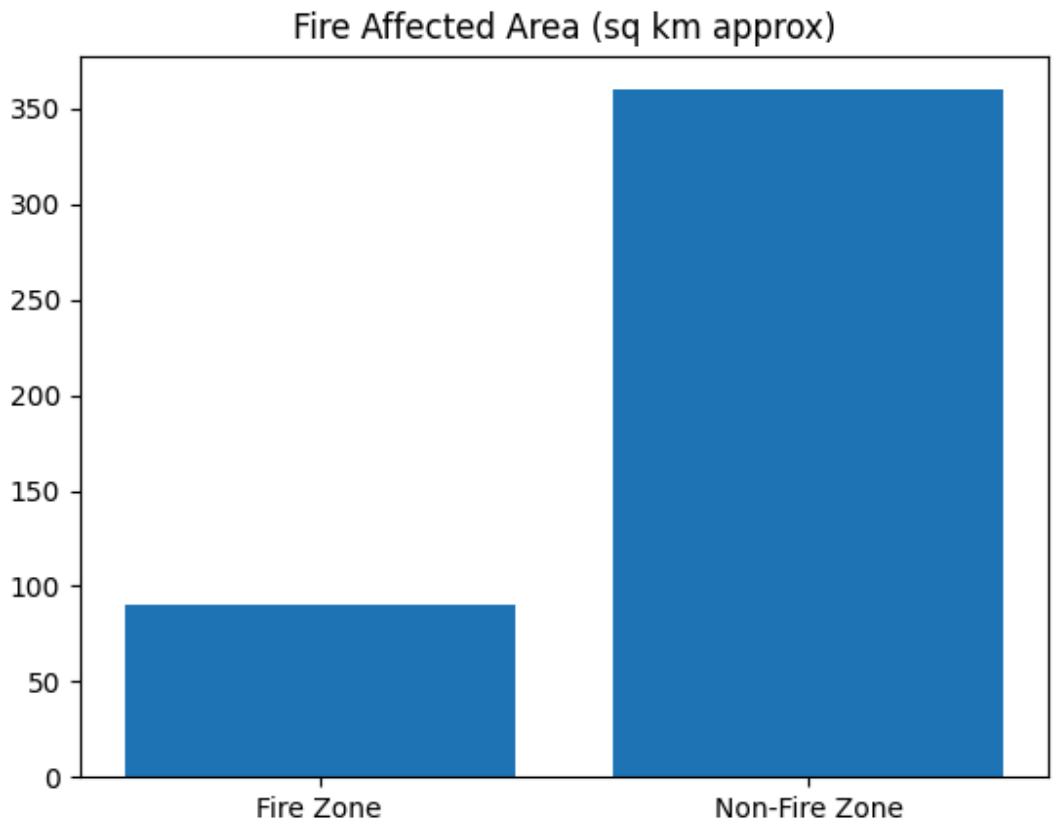
Represents the proportion of opencast and underground mining.
Opencast mining contributes more due to higher productivity and lower cost.
Significance: Useful for comparing mining methods and understanding cost-efficiency in coal extraction.

📊 3. Coking Coal Distribution



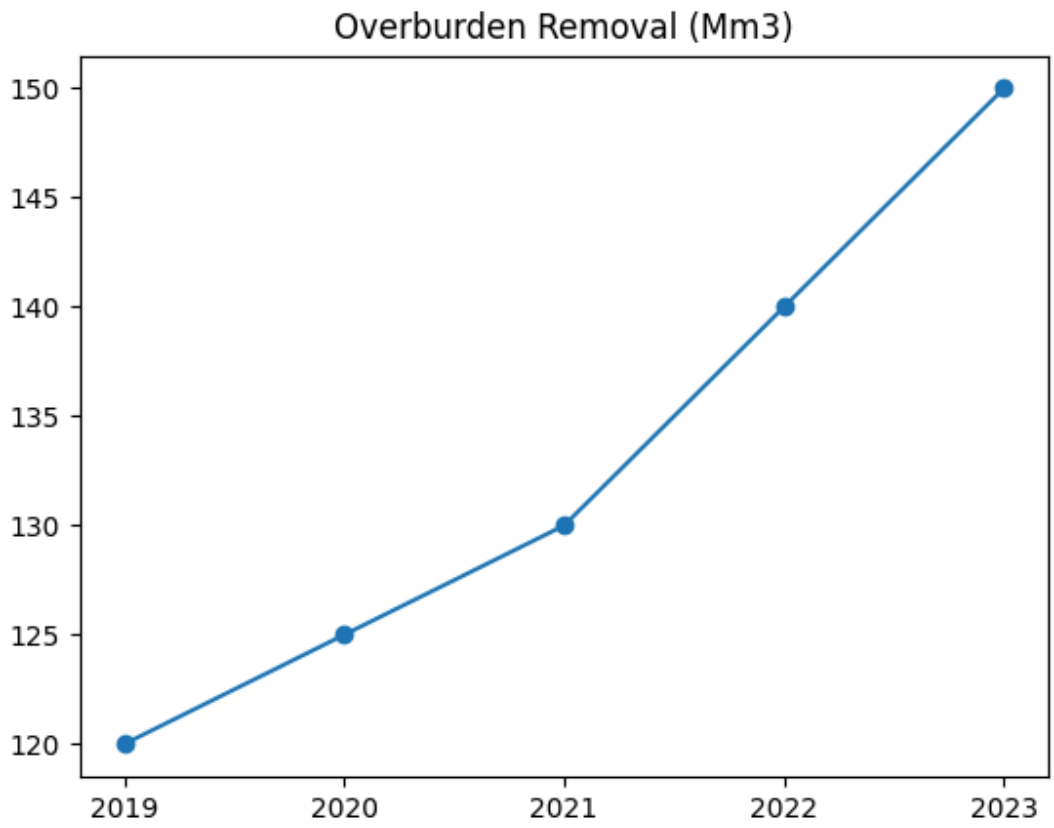
Shows the proportion of prime, medium, and semi-coking coal in Jharia.
A high percentage of prime coking coal makes this coalfield unique.
Significance: Critical for steel production and coal beneficiation planning.

📊 4. Fire-Affected Area



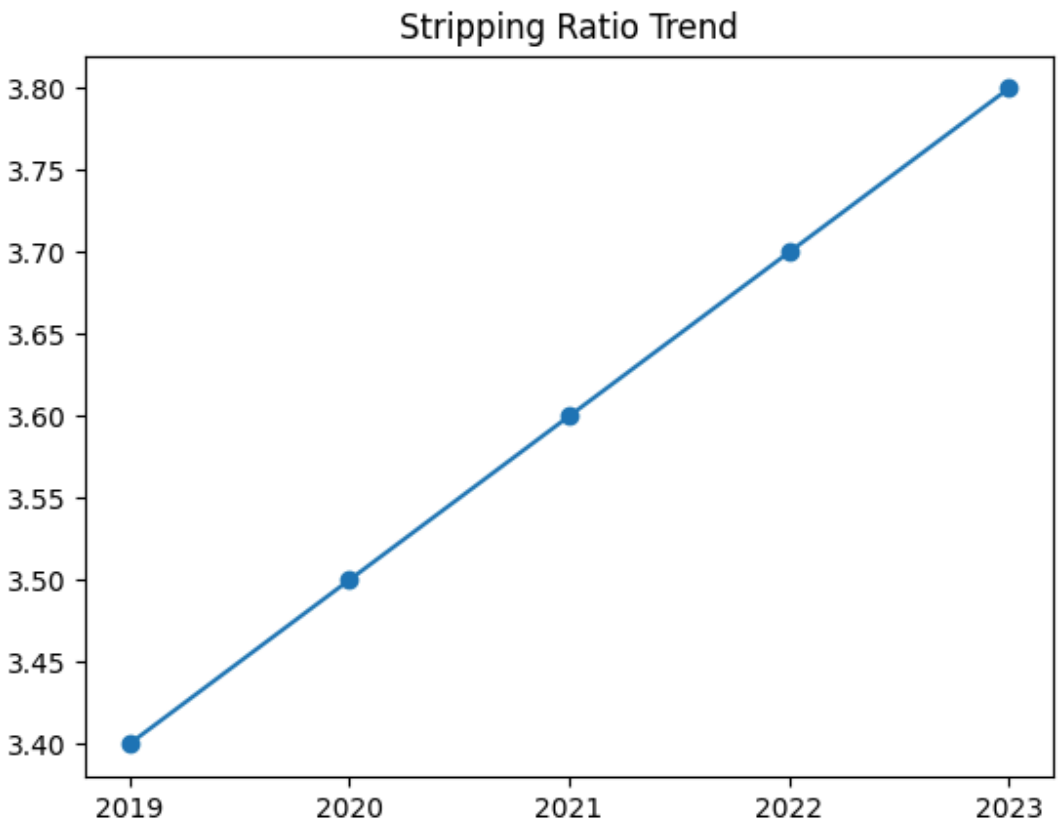
This chart highlights the area affected by underground coal fires.
A significant portion of land is impacted due to long-burning fires.
Significance: Important for safety planning, land reclamation, and environmental management.

📊 5. Overburden Removal (Mm³)



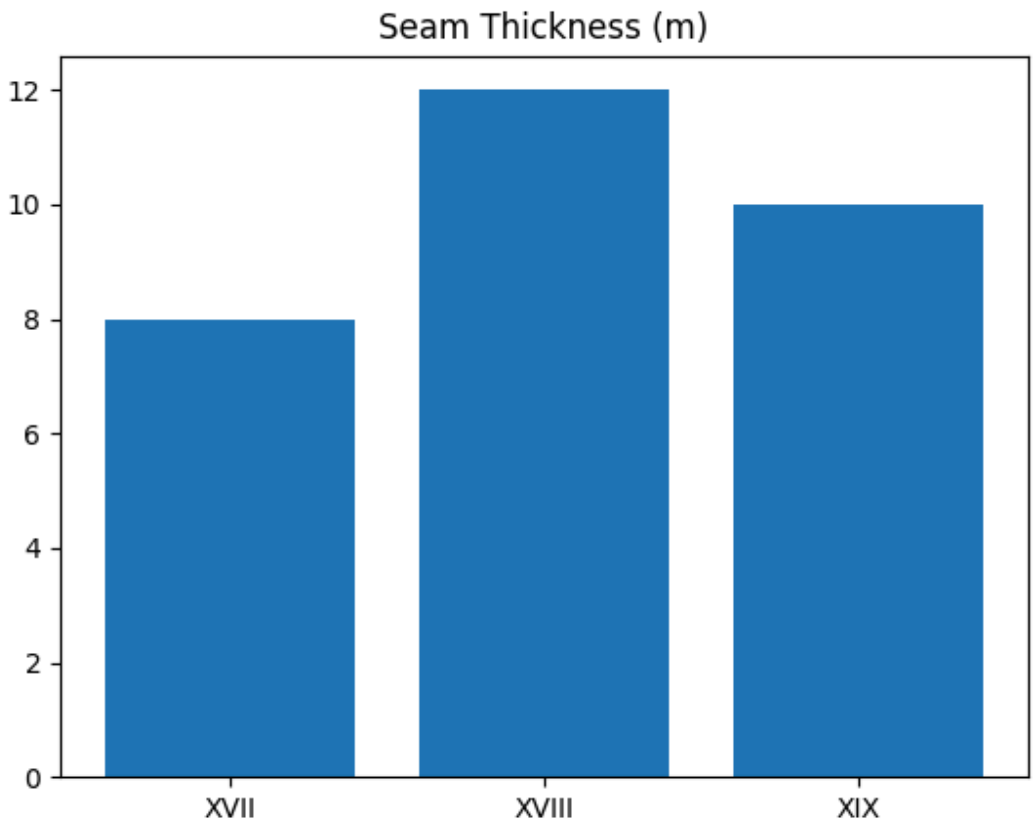
Displays the quantity of overburden removed to access coal seams.
Increasing values indicate expansion of opencast mining.
Significance: Helps in calculating stripping ratio and planning dump design.

📊 6. Stripping Ratio Trend



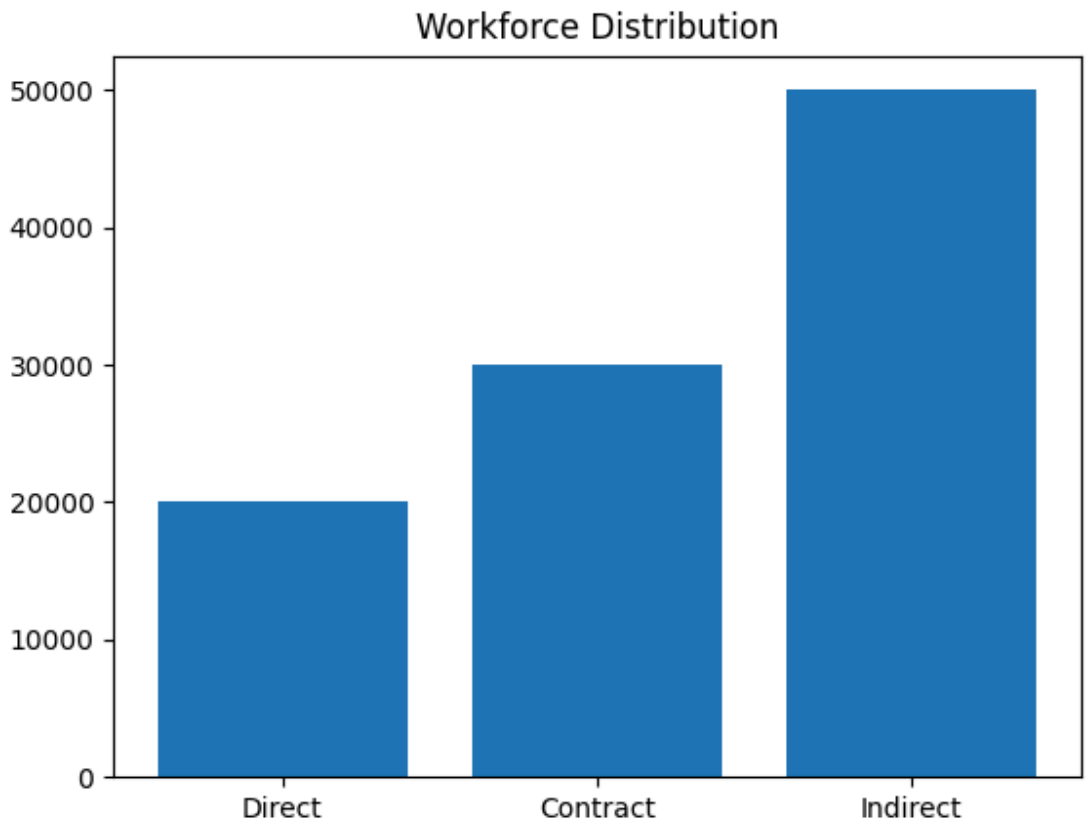
Shows the relationship between overburden and coal production.
A gradual rise indicates increasing mining depth and cost.
Significance: Key parameter for economic feasibility and mine planning.

📊 7. Seam Thickness (m)



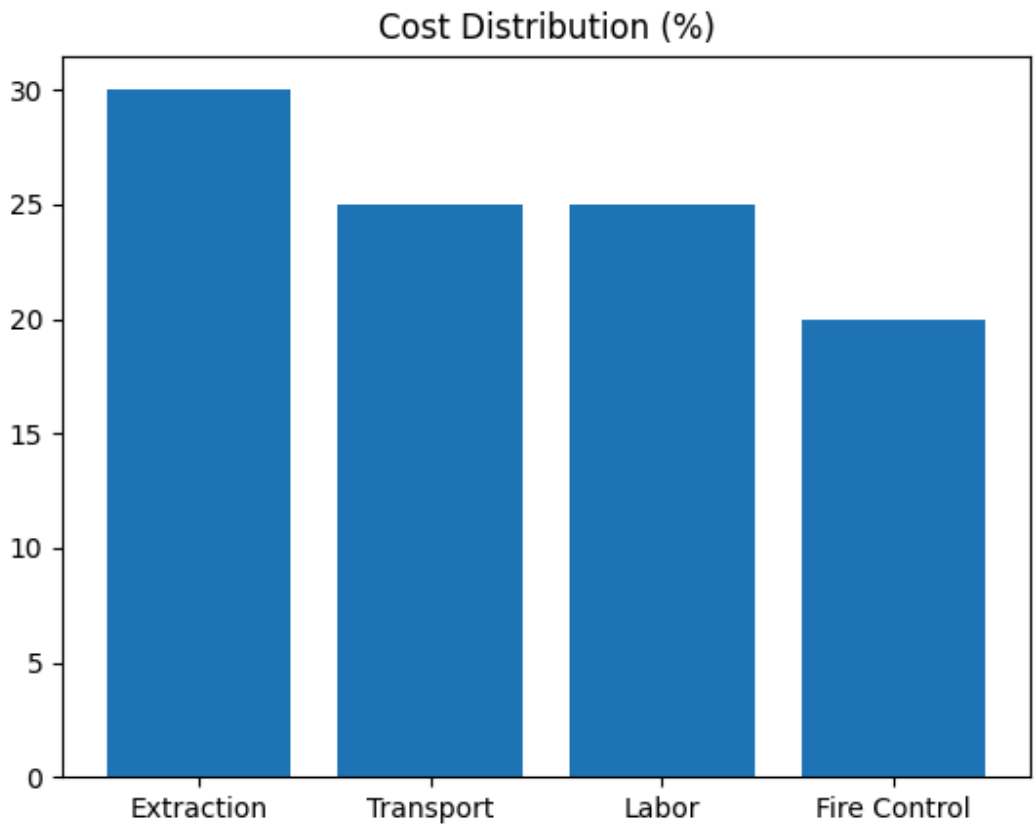
Represents thickness of major coal seams in Jharia.
Thicker seams allow more efficient and economical extraction.
Significance: Important for selecting mining method and equipment.

📊 8. Workforce Distribution



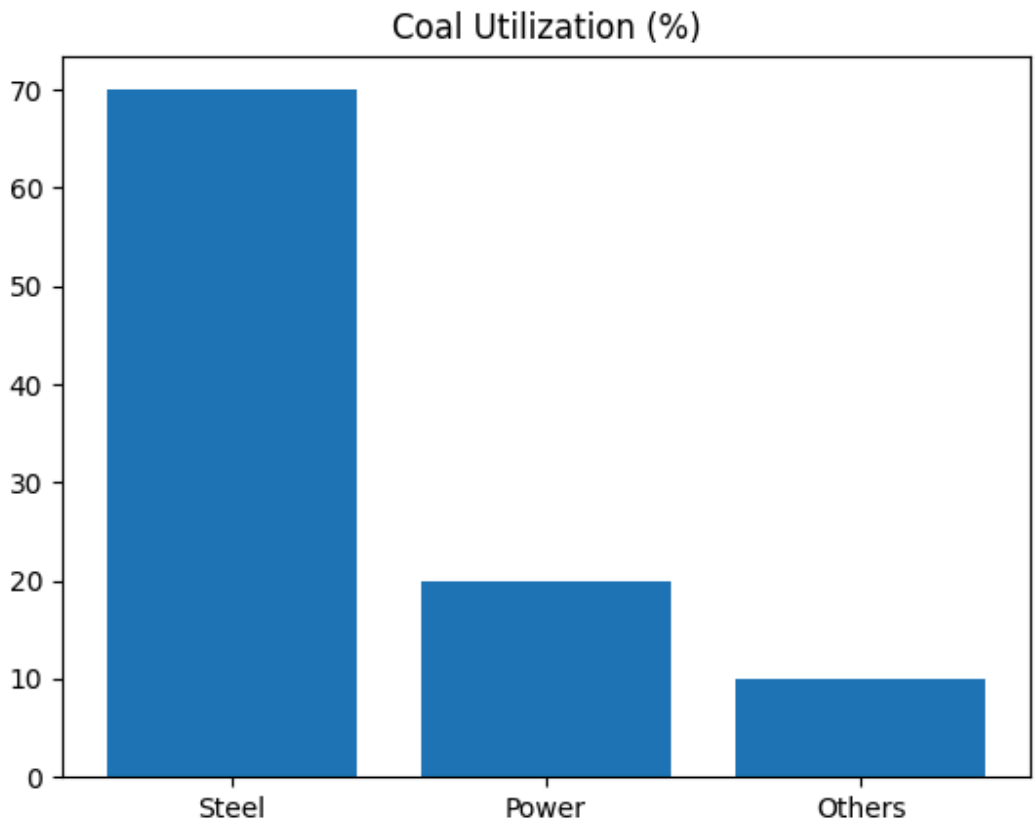
Shows distribution of manpower across direct, contract, and indirect categories.
Highlights large employment generated by mining operations.
Significance: Useful for manpower planning and socio-economic analysis.

📊 9. Cost Distribution (%)



Displays cost components including extraction, transport, labor, and fire control.
Fire control adds a unique cost factor in Jharia.
Significance: Helps in cost optimization and understanding economic challenges.

📊 10. Coal Utilization (%)



Shows distribution of coal usage across steel, power, and other sectors.
Steel industry consumes the majority due to coking coal demand.
Significance: Highlights the strategic importance of Jharia in India's steel sector.